①

EXERCISE 4 Transcribe Publication (Extra Science)

SQL Joins

① Retrieve all employees and assigned projects using an inner join.

→ Return EmployeeID, LastName, Department, Salary, ProjectID, ProjectName, Budget, status

```
Select A.EmployeeID,
       A.LastName,
       A.Department,
       A.Salary,
       B.ProjectID,
       B.ProjectName,
       B.Budget,
       B.Status
From brightlearn.database.excerise_4_Employees as A
inner join brightlearn.database.excerise_4_projects as B
on A.EmployeeID = B.EmployeeID;
```

Employees ID	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	Deo	IT	70000	101	AI Development	100000	Completed
2	Smith	HR	60000	102	Employee Training	50000	ongoing
3	Deo	IT	70000	103	Cybersecurity Audit	75000	Pending
4	Johnson	Finance	75000	104	Finance Analysis	90000	ongoing
5	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Clark	Finance	80000	106	Risk Management	80000	Pending

② EXERCISE 4 Thulasiram Buttiela
(Data Science)

② Retrieve all employees and their assigned projects including employees who have no projects using a Left join.

* Return EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Deo	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	ongoing
4	David	Brown	IT	72000	null	null	null	null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	50000	pending
1	John	Deo	IT	7000	101	AI Development	100000	Complete

SQL Script

```
Select A. EmployeeID,
       A. FirstName,
       A. LastName,
       A. Department,
       A. Salary,
       B. ProjectID,
       B. ProjectName,
       B. Budget,
       B. Status
from brightlearn.database.exercise_4_employees as A
Left join brightlearn.database.exercise_4_projects AS
on A.EmployeeID = B.EmployeeID
```

③ Exercise 4 Thulasigure Ruthelan
(Data Science)

③ Retrieve all projects and their assigned Employees, including Projects that have no employees using a Right Join
* Run: ProjectID, ProjectName, Budget, Status, EmployeeID, LastName, Department, Salary

Select A. ProjectID,
A. ProjectName,
A. Budget,
A. Status,
B. FirstName,
B. LastName,
B. Department,
B. Salary

From brightlearn.~~database~~. excerise_4_employees as B
right join brightlearn. database. Excerise_4_~~employ~~ projects as A
on B. EmployeeID = A. EmployeeID.

Project ID	Project Name	Budget	Status	EmployeeID	First Name	Last Name	Department	Salary
101	AI Development	100 000	Completed	1	John	Doe	IT	70 000
102	Employee Training	50 000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity Audit	75 000	Pending	1	John	Doe	IT	70 000
104	Financial Analysis	90 000	Ongoing	3	bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65 000
106	Risk Management	86000	Pending	6	Michael	Clark	Finance	80 000

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~~Exer~~ Exercise 4

Thulasiswe Pethelezi
(Data Science)

④ Retrieve all employees and projects, including those without a match in either table using a Full Outer Join

* Return EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, status

```
select A.EmployeeID,
       A.FirstName,
       A.LastName,
       A.Department,
       A.Salary,
       B.ProjectID,
       B.ProjectName,
       B.Budget,
       B.status,
from brightlearn.databases.exercise_4_employees AS A
Full outer join brightlearn.databases.exercise_4_projects
AS B
On A.EmployeeID = B.EmployeeID.
```

EmployeeID	First Name	Last Name	Department	Salary	ProjectID	Project Name	Budget	Status
3	Bob ~~David~~	Johnson	Finance	75000	104	Financial analysis	90000	ongoing
4	David ~~Williams~~	Brown	IT	72000	null	null	null	null
6	Michael ~~Alice~~	Clark	Finance	80000	106	Risk Management	50000	Pending
2	Alice ~~Emma~~	Smith	HR	60000	102	Employee Training	58000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	completed
1	John	Doe	IT	70000	103	Cyber Security Audit	75000	Pending
1	John	Doe	IT	70000	101	AI Development	100000	Completed

⑤ union and union

⑤ Retrive a list of all unique cities where employees are located and project statuses

* Return: Location (Rename the column to location using an alias)

```
Select distinct(city) as locations
From brightlearn.databases.exercise_4_Employees
Union
Select status
From brightlearn.databases.exercise_4_projects;
```

??

⑤

	Location
1	London
2	New York
3	Los Angeles
4	Sydney
5	Toronto

⑥

	location
1	New York
2	Los Angeles
3	Toronto
4	London
5	Sydney
6	New York

Project status below the same column → it confusing.

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```
Select city as location
From brightlearn.databases.exercise_4_Employees
Union ALL
Select status
From brightlearn.databases.exercises_4_Projects
```

(6) Filtering statements

Thulasisure Bulbolet
(Data Science)

(7) Retrieve employees who earn 70 000

* Return EmployeeID, FirstName, LastName, Department, Salary

```
Select EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
From brightlearn.database.Exercise 4_Employees
Where Salary > 70 000;
```

Employee ID	First Name	Last Name	Department	Salary
3	John	Doe	IT	7
3	Bob	Johnson	Finance	
4	David	Brown	IT	
6	Michael	Clark	Finance	

Employee ID	First Name	Last Name	Department	Salary
3	BoB	Johnson	Finance	15000
4	David	Brown	IT	12000
6	Michael	Clark	finance	80000

(7) Exercise 4 — Thulasizwe Buthelezi Data Science.

8. Retrieve employees working in IT or Finance departments

→ Return: EmployeeID, FirstName, LastName, Department, Salary

```
Select EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
From brightlearn.databases.Exercise_4_employees
Where Department in ('IT', 'Finance');
```

Employee ID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9. Retrieve projects that are not yet completed

Return: ProjectID, ProjectName, Budget, status.

```
Select ProjectID,
       ProjectName,
       Budget,
       Status
From brightlearn.databases.Exercise_4_Project
Where status not in ('Completed');
```

Project ID	Project Name	Budget	Status
102	Employee Training	50000	ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	ongoing
106	Risk Management	80000	Pending

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Exercise 4

Thulanizwe Bulelani
Data Science

10. Retrieve projects that have a budget greater than 70 000 and are not completed.
* Return: ProjectID, ProjectName, Budget, Status

```
Select ProjectID,
       ProjectName,
       Budget,
       Status
From bright learn.database.exercise_4_Projects
Where Budget > 70 000 and Status not in ('completed');
```

ProjectID	ProjectName	Budget	Status
103	Cyber Security Audit	75 000	Pending
104	Financial Analysis	90 000	ongoing
106	Risk Management	80 000	Pending

11. Retrieve employees from New York or Toronto, ordered by Salary in descending order.
* Return: EmployeeID, FirstName, LastName, Department, Salary, City

```
Select EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       City
From Bright learn.database.Exercise_4_Projects
where City in ('New York', 'Toronto')
order by Salary desc;
```

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Exercise 11

Thulasiram Butterfly
Data Science

11. Retrieve employees from New York or Toronto, order by Salary in descending order

Return: EmployeeID, FirstName, LastName, Department, Salary, City

~~Select EmployeeID, FirstName, LastName, Department,~~

```
Select EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       City
From brightlearn.databases.Exercise_4_Employees
Where City in ('New York', 'Toronto')
order by Salary desc;
```

Employee ID	First Name	Last Name	Department	Salary	City
6	Michael	Clark	Finance	80 000	New York
3	Bob	Johnson	Finance	15 000	Toronto
1	John	Doe	IT	10 000	New York

12. Retrieve the top 3 highest-paid employees

Return: EmployeeID, FirstName, LastName, Department, Salary

```
Select EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
From brightlearn.databases.Exercise_4_Employees
order by Salary desc
Limit 3;
```

Employee ID	First Name	Last Name	Salary	Department
6	Michael	Clark	80 000	Finance
3	Bob	Johnson	15 000	Finance
4	David	Brown	12 000	IT

(10) Exercise 4

Thulasiram Butterai
Data Science

(4) Aggrege Functions

13. Find the total Salary department, sorted in descending Order

Return: Department, TotalSalary(Rename(Sum(Salary) as TotalSalary

```
Select Department,
    sum(salary) as Total_Salary

From brightlearn.database.exercise_4_Employees
group by Department
~~Order by Salary desc~~;
order by Total_Salary;
```

Department	Total Salary
Finance	155000
IT	142000
Sales	65000
HR	6000

14. Find the average salary per city, but only include cities where the average salary is greater than 65 000

Return: City, AverageSalary, (Rename(Avg(Salary) as Average Salary

```
Select City,
    Avg(Salary) as Average_Salary

From brightlearn.databases.Exercise_4_Employees
where Average_Salary > 65000
group by City
```

City	Average Salary

City	Average_Salary
New York	75000
Toronto	75000
London	72000

(18)

Exercise 4 Thulasizwe Butholezi
Data Science

15. Count the number of employees per department, Including only departments with more than 1 employee

* Return: Department, EmployeeCount (Rename (Count (EmployeeID) as EmployeeCount

~~Select Department,~~
~~Count (EmployeeID) as EmployeeCount~~
~~from brightlearn.databases.Exercise_4_Employees~~
~~Where EmployeeCount > 1~~
~~group by Department;~~

Select Department,
Count (EmployeeID) as EmployeeCount;

From brightlearn.databases.Exercise_4_Employees
Group by Department
Having Count(EmployeeID) > 1;

Department	Employee count
IT	2
Finance	2

16. Retrive the number of projects per status but only include statuses with at least 2 Projects- Return: Status, ProjectCounts)

Select Status,
Count(ProjectID) as ProjectCount
From brightlearn.databases.Exercises_4_Projects
Group by Status
Having Count(ProjectID) ≥ 2;

Status	ProjectCount
Completed	2
ongoing	2
Pending	2.

(17) Exercise 4

Thulasiram Bathula
Data Science

17. Retrieve the total projects budget per employee, but only for Employees who are managing projects worth more than 150 000

Return: EmployeeID, FirstName, LastName, TotalBudget

```
Select A.EmployeeID,
       B.FirstName,
       B.LastNam,
       Sum(A.Budget) as Totalbudget
From brightlearn.databases.exercise_4_Projects as A
left join brightlearn.dlabases.exercise_4_Employees as B
on A.EmployeeID = B.EmployeeID,
group by A.EmployeeID, B.FirstName, B.LastName.
having Sum(A.Budget) > 150000
```

Employee ID	First Name	Last Name	Total budget
1	John	Deo	175000